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AI/ML Intern

MODULE : 2

BODY MEASUREMENTS USING POSE DETECTION

CODE USED FOR DEMONSTRATION

pd.py > ...

```python
import cv2
import mediapipe as mp
import math
from tkinter import Tk
from tkinter.filedialog import askopenfilename

real_height_cm=float(input("Enter customer height in cm: "))

Tk().withdraw()

image_path=askopenfilename(
title="Select Customer Image",
filetypes=[("Image Files","*.jpg *.jpeg *.png")]
)

image=cv2.imread(image_path)

if image is None:
    print("Image not loaded.")
    exit()

image=cv2.resize(image,(900,700))

mp_pose=mp.solutions.pose
mp_draw=mp.solutions.drawing_utils

pose=mp_pose.Pose(
static_image_mode=True,
model_complexity=2,
min_detection_confidence=0.7
)

rgb=cv2.cvtColor(image,cv2.COLOR_BGR2RGB)
results=pose.process(rgb)

def distance(p1,p2):
    return math.sqrt((p2[0]-p1[0])**2+(p2[1]-p1[1])**2)

def point(lm,w,h):
    return (lm.x*w,lm.y*h)

def draw_text(text,y,color):
```

pd.py > ...

```python
def draw_text(text,y,color):
    cv2.putText(
    image,
    text,
    (20,y),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    color,
    2
    )

if results.pose_landmarks:

    landmarks=results.pose_landmarks.landmark
    h,w,_=image.shape

    mp_draw.draw_landmarks(
    image,
    results.pose_landmarks,
    mp_pose.POSE_CONNECTIONS
    )

    for lm in landmarks:
        x=int(lm.x*w)
        y=int(lm.y*h)
        cv2.circle(image,(x,y),4,(0,255,0),-1)

    p={
    "ls":landmarks[mp_pose.PoseLandmark.LEFT_SHOULDER],
    "rs":landmarks[mp_pose.PoseLandmark.RIGHT_SHOULDER],
    "lh":landmarks[mp_pose.PoseLandmark.LEFT_HIP],
    "rh":landmarks[mp_pose.PoseLandmark.RIGHT_HIP],
    "le":landmarks[mp_pose.PoseLandmark.LEFT_ELBOW],
    "lw":landmarks[mp_pose.PoseLandmark.LEFT_WRIST],
    "lk":landmarks[mp_pose.PoseLandmark.LEFT_KNEE],
    "la":landmarks[mp_pose.PoseLandmark.LEFT_ANKLE],
    "ra":landmarks[mp_pose.PoseLandmark.RIGHT_ANKLE],
    "nose":landmarks[mp_pose.PoseLandmark.NOSE]
    }

    shoulder_pixels=distance(
    point(p["ls"],w,h),
```

pd.py > ...

```python
    point(p["ls"],w,h),
    point(p["rs"],w,h)
    )

    waist_pixels=distance(
    point(p["lh"],w,h),
    point(p["rh"],w,h)
    )

    sleeve_pixels=distance(
    point(p["ls"],w,h),
    point(p["le"],w,h)
    )+distance(
    point(p["le"],w,h),
    point(p["lw"],w,h)
    )

    torso_pixels=distance(
    point(p["ls"],w,h),
    point(p["lh"],w,h)
    )

    leg_pixels=distance(
    point(p["lh"],w,h),
    point(p["lk"],w,h)
    )+distance(
    point(p["lk"],w,h),
    point(p["la"],w,h)
    )

    avg_ankle_y=((p["la"].y+p["ra"].y)/2)*h

    body_height_pixels=abs((p["nose"].y*h)-avg_ankle_y)

    scale_factor=real_height_cm/body_height_pixels

    measurements={
    "Shoulder Width":shoulder_pixels*scale_factor,
    "Waist Width":waist_pixels*scale_factor,
    "Sleeve Length":sleeve_pixels*scale_factor,
    "Torso Length":torso_pixels*scale_factor,
    "Leg Length":leg_pixels*scale_factor
```

```python
    "Leg Length":leg_pixels*scale_factor
    }

    lines=[
    (p["ls"],p["rs"],(255,0,255)),
    (p["lh"],p["rh"],(0,255,255)),
    (p["ls"],p["le"],(0,0,255)),
    (p["le"],p["lw"],(0,0,255)),
    (p["ls"],p["lh"],(255,255,0)),
    (p["lh"],p["lk"],(0,255,0)),
    (p["lk"],p["la"],(0,255,0))
    ]

    for p1,p2,color in lines:
        cv2.line(
        image,
        (int(p1.x*w),int(p1.y*h)),
        (int(p2.x*w),int(p2.y*h)),
        color,
        3
        )

    colors=[
    (255,0,255),
    (0,255,255),
    (0,0,255),
    (255,255,0),
    (0,255,0)
    ]

    for i,(name,value) in enumerate(measurements.items()):
        draw_text(
        f"{name}: {value:.1f} cm",
        40+(i*40),
        colors[i]
        )

    print("\nMeasurements:\n")

    for name,value in measurements.items():
        print(f"{name}: {value:.2f} cm")
```

```python
    for name,value in measurements.items():
        print(f"{name}: {value:.2f} cm")

    cv2.imshow("RARITONE Body Measurement System",image)

    cv2.waitKey(0)
    cv2.destroyAllWindows()

else:
    print("Pose not detected properly.")
```

This code is used to measure different body lengths of a human body to get the perfect size for the customer.

THE OUTPUT FOR THIS CODE IS:


```
(venv) PS C:\Users\Akshith Kalvakota\Desktop\pose detection> python pd.py
Enter customer height in cm: 175
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.

Measurements:

Shoulder Width : 52.24 cm
Waist Width    : 28.20 cm
Sleeve Length  : 62.60 cm
Torso Length   : 64.64 cm
Leg Length     : 89.84 cm
(venv) PS C:\Users\Akshith Kalvakota\Desktop\pose detection>
```

This code provides us the:

- Shoulder width
- Waist Width
- Sleeve Length
- Torso Length
- Leg Length

But first the user has to mention his/her height in cm so that conversion formulae can be used to convert the pixels length into actual human used measurements like centimeters.

LOGIC BEHIND IT:

1) The height of the user is taken.

2) Then the pose detection using MediaPipe detects the landmarks on the body and assigns a coordinate(x,y) to each landmark.

3) The distance between different landmarks is calculated to get the requirement measurements. For example, for shoulder width we take the distance between the left shoulder and the right shoulder.

The distance is taken using the Euclidean distance formula:

For (x,y) :

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

4) Now the distance taken is in pixels, so we use the real height of the user in cm to calculate the scale factor:

$$\text{Scale Factor} = \frac{\text{Real Height in cm}}{\text{Detected Height in Pixels}}$$

5) Now we convert each body measurement into cm using, for example:

SHOULDER WIDTH (CM) = SHOULDER PIXELS X SCALE FACTOR

And that is how we get the final measurements in cm.